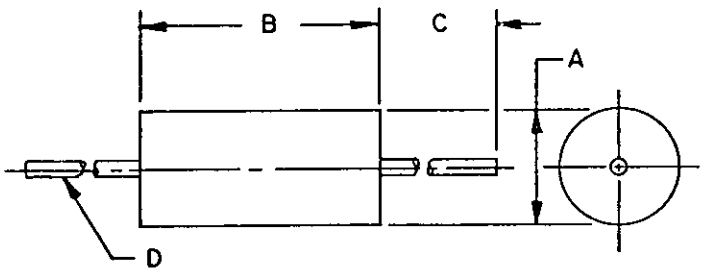
MILITARY SPECIFICATION SHEET

COILS, FIXED, RADIOFREQUENCY, MOLDED, MICRO-MINIATURE, MAGNETICALLY SHIELDED (FERRITE CORE-FERRITE SLEEVE), ESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the coils described herein shall consist of this specification and the latest issue of MIL-C-39010.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.152 (3.86)	.172 (4.37)
B	.390 (9.91)	.430 (10.92)
C	1.250 (31.75)	1.626 (41.30)
D	.023 (.58)	.027 (.69)

NOTE: Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Material: Ferrite core - ferrite sleeve.

Weight: 1.00 gram maximum.

Operating temperature range: -55°C to +105°C.

Dielectric withstanding voltage: Method 301 of MIL-STD-202 test voltage: 1,000 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202 (70,000 feet), test voltage 200 volts rms.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Inductance tolerance: See table I.

FSC 5950

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:30Z
Check the source to verify that this is the current version before use.

Q values: See table I.

Self-resonant frequency: See table I.

Percent coupling: 3 percent maximum.

DC resistance: See table I.

Temperature rise: 15°C.

Terminal pull: 5 pounds.

Part number: M39010/03 (dash number from table I).

Revision letters are not used to denote changes due to the extensiveness of the changes.

TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/	Incremental current
A150**	15	5,10	45	2.5	49	.80	315.0	250
A160**	16	5	45	2.5	45	.89	300.0	235
A180**	18	5,10	45	2.5	45	.89	300.0	235
A200**	20	5	45	2.5	41	.96	290.0	220
A220**	22	5,10	45	2.5	41	.96	290.0	220
A240**	24	5	45	2.5	38	1.19	260.0	200
A270**	27	5,10	45	2.5	38	1.19	260.0	200
A300**	30	5	45	2.5	34	1.37	240.0	190
A330**	33	5,10	45	2.5	34	1.37	240.0	190
A360**	36	5	45	2.5	29	1.93	205.0	180
A390**	39	5,10	50	2.5	29	1.93	205.0	180
A430**	43	5	50	2.5	27	2.11	195.0	175
A470**	47	5,10	50	2.5	27	2.11	195.0	175
A510**	51	5	50	2.5	25	2.23	190.0	160
A560**	56	5,10	50	2.5	25	2.23	190.0	160
A620**	62	5	50	2.5	21	2.70	170.0	150
A680**	68	5,10	50	2.5	21	2.70	170.0	150
A750**	75	5	50	2.5	10.5	2.44	180.0	140
A820**	82	5,10	50	2.5	10.5	2.44	180.0	140
A910**	91	5	50	2.5	10.0	3.12	160.0	120

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:30Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/	Incremental current
A101**	100	5,10	50	2.5	10.0	3.12	160.0	120
A111**	110	5	50	.79	9.7	3.60	150.0	95
A121**	120	5,10	55	.79	9.7	3.60	150.0	95
A131**	130	5	55	.79	8.5	4.10	140.0	90
A151**	150	5,10	55	.79	8.5	4.10	140.0	90
A161**	160	5	55	.79	8.0	4.40	135.0	85
A181**	180	5,10	55	.79	8.0	4.40	135.0	85
A201**	200	5	55	.79	7.5	5.00	125.0	80
A221**	220	5,10	55	.79	7.5	5.00	125.0	80
A241**	240	5	55	.79	7.0	5.80	115.0	70
A271**	270	5,10	55	.79	7.0	5.80	115.0	70
A301**	300	5	55	.79	6.5	6.40	110.0	65
A331**	330	5,10	55	.79	6.5	6.40	110.0	65
A361**	360	5	55	.79	6.2	7.40	105.0	60
A391**	390	5,10	60	.79	6.2	7.40	105.0	60
A431**	430	5	60	.79	5.7	9.50	92.0	58
A471**	470	5,10	60	.79	5.7	9.50	92.0	58
A511**	510	5	60	.79	4.7	10.5	90.0	55
A561**	560	5,10	60	.79	4.7	10.5	90.0	55
A621**	620	5	60	.79	4.5	11.8	80.0	50
A681**	680	5,10	60	.79	4.5	11.8	80.0	50
A751**	750	5	60	.79	4.2	13.0	80.0	45
A821**	820	5,10	60	.79	4.2	13.0	80.0	45
A911**	910	5	60	.79	3.8	17.5	70.0	40
A102**	1,000	5,10	60	.79	3.8	17.5	70.0	40
A112**	1,100	5	45	.25	1.5	22.1	60.0	35
A122**	1,200	5,10	45	.25	1.5	22.1	60.0	35
A132**	1,300	5	45	.25	1.2	26.5	55.0	33

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:30Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/	Incremental current
A152**	1,500	5,10	45	.25	1.2	26.5	55.0	33
A162**	1,600	5	45	.25	1.0	29.9	50.0	30
A182**	1,800	5,10	45	.25	1.0	29.9	50.0	30
A202**	2,000	5	45	.25	.97	33.8	50.0	27
A222**	2,200	5,10	45	.25	.97	33.8	50.0	27
A242**	2,400	5	45	.25	.92	47.3	40.0	25
A272**	2,700	5,10	45	.25	.92	47.3	40.0	25
A302**	3,000	5	45	.25	.84	53.0	40.0	22
A332**	3,300	5,10	45	.25	.84	53.0	40.0	22
A362**	3,600	5	45	.25	.80	73.8	35.0	20
A392**	3,900	5,10	45	.25	.80	73.8	35.0	20
A432**	4,300	5	45	.25	.74	81.6	31.0	19
A472**	4,700	5,10	45	.25	.74	81.6	31.0	19
A512**	5,100	5	44	.25	.73	98.9	28.0	17
A562**	5,600	5,10	44	.25	.73	98.9	28.0	17
A622**	6,200	5	40	.25	.66	111.0	27.0	16
A682**	6,800	5,10	40	.25	.66	111.0	27.0	16
A752**	7,500	5	40	.25	.54	119.0	26.0	15
A822**	8,200	5,10	40	.25	.54	119.0	26.0	15
A912**	9,100	5	40	.25	.47	137.0	24.0	14
A103**	10,000	5,10	40	.25	.47	137.0	24.0	14
A113**	11,000	5	30	.079	.33	143.0	23.0	13
A123**	12,000	5,10	30	.079	.33	143.0	23.0	13
A133**	13,000	5	30	.079	.29	157.0	22.0	12
A153**	15,000	5,10	30	.079	.29	157.0	22.0	12
A163**	16,000	5	30	.079	.28	175.0	21.0	10
A183**	18,000	5,10	30	.079	.28	175.0	21.0	10
A203**	20,000	5	27	.079	.25	274.0	17.0	9
A223**	22,000	5,10	27	.079	.25	274.0	17.0	9

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:30Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/	Incremental current
A243**	24,000	5	27	.079	.21	308.0	16.0	8
A273**	27,000	5,10	27	.079	.21	308.0	16.0	8
A303**	30,000	5	27	.079	.19	343.0	15.0	7.5
A333**	33,000	5,10	27	.079	.19	343.0	15.0	7.5
A363**	36,000	5	27	.079	.17	376.0	15.0	6.0
A393**	39,000	5,10	27	.079	.17	376.0	15.0	6.0
A433**	43,000	5	23	.079	.16	473.0	13.0	5.5
A473**	47,000	5,10	23	.079	.16	473.0	13.0	5.5
A513**	51,000	5	23	.079	.14	512.0	13.0	5.0
A563**	56,000	5,10	23	.079	.14	512.0	13.0	5.0
A623**	62,000	5	23	.079	.13	580.0	12.0	4.0
A683**	68,000	5,10	23	.079	.13	580.0	12.0	4.0
A753**	75,000	5	21	.079	.12	618.0	11.0	3.5
A823**	82,000	5,10	21	.079	.12	618.0	11.0	3.5
A913**	91,000	5	18	.079	.11	678.0	11.0	3.0
A104**	100,000	5,10	18	.079	.11	678.0	11.0	3.0

1/ The complete dash number will include two additional letters (indicated by **). The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the failure rate symbol (e.g., M, P, R, S) and will be added to the end of the dash number.

2/ The rated dc current is based on 90°C ambient temperature with a 15°C rise.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:30Z
Check the source to verify that this is the current version before use.

TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from initial measurement		Allowable % from specified minimum value in electrical characteristics (initial) table	
	Inductance	DC resistance	Self-resonant frequency	Q
	Percent			
Qualification inspection				
Group II	±5	---	---	-10
Group IV	±10	±(5%+.001 ohm)	1/	-20
Group V	±5	±(2%+.001 ohm)	1/	-20
Quality conformance inspection				
Group C				
Subgroup 1	±5	±(2%+.001 ohm)	1/	-20
Subgroup 3	±5	---	---	-10
Subgroup 4	±10	±(5%+.001 ohm)	1/	-20

1/ The self-resonant frequency shall be not less than the value specified in table I.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - AS, CG, MC, SH
Air Force - 19

Agent:
DLA-ES

Preparing activity:
Army - ER

(Project 5950-0603-3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:30Z
Check the source to verify that this is the current version before use.